

Benefits & Costs of Friendship Networks

Considering the Evolutionary Benefits and Costs of Friendship Networks

Grace Vieth, Ryan T. Dobson, Jeffrey Simpson, Steven W. Gangestad

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Abstract

Recent efforts have sought to redress the relative paucity of theoretical work on the adaptive benefits that led friendships to be fundamentally important close human relationships (e.g., viewing investment in them as stake-based). Still, relatively little work has specifically addressed adaptations that function to developing, maintaining, and regulating friendship networks, generally consisting of multiple friends. This paper explores this domain. Following other scholars, we observe that a key quality of friendships that maintains stake-based investment in them is irreplaceability. Adding a friend to one's friendship network may have marginal net benefits. Yet the added friend may not only be less irreplaceable than others in the network; the added friend may reduce the irreplaceability of other friends in the network, thereby potentially weakening stake-based investment crucial to friendship. We discuss multiple adaptive problems that arise and potential solutions to them, suggestive of adaptations that people possess for the function of developing, maintaining, and regulating friendship networks. Future directions for potentially fruitful conceptual and empirical exploration are discussed, as are potential domains of application to understanding friendship more broadly.

1. The Importance of Friends and Friendship Networks

People's most important, close relationships with other people come in a variety of forms. Kinship is one important form, with relations between parents and their offspring, as well as sibships, especially so (Hamilton, 1964). Pair-bonds are also central (Chapais, 2017). Another form is friendship. Friendship has been defined as “a voluntary, personal relationship, typically providing intimacy and assistance, in which two parties like one another and seek each other's company” (Fehr, 1996, p. 20), which, though accurate, understates the centrality of friendships to people's lives. Not only are friends sources of happiness and personal fulfillment (e.g., Chopik, 2017); people who report having satisfying friendships are psychologically and physically healthier: They have a smaller risk of developing chronic illnesses, have better cognitive functioning in older age, and even live longer (Demir et al., 2013; Béland et al., 2005; Chopik, 2017; Holt-Lunstad et al., 2010; Holt-Lunstad, 2018). Not surprisingly, then, when people are asked to identify the most important elements of their lives, friendships fall near the top (see, e.g., Dunbar, 2022). In one U.S. survey, people on average ranked friendships as more vital to a fulfilling life than wealth, marriage, or having children (Goddard, 2023).

Because they are distinct from kinship and romantic partnerships, friendships have some unique features. Unlike kinship, they are voluntary. In comparison to pair-bonds, they are less constrained by social structures and norms surrounding institutions (e.g., marriage; Becker et al., 2009; Vieth et al., 2025). Nevertheless, friendships are stable and enduring for many adults (Pennington et al., 2024). Almost all people develop at least one good friendship. Most invest in multiple friendships; people in the U.S. report average roughly four to five close friends (Pennington et al., 2024; Thompson et al., 2024; see also Dunbar, 2018, 2022).

Tooby and Cosmides (1996) offered key insights of how non-kin relationships characterized by “deep engagement”—notably, close friendships—may have evolved. Their ideas relate to Aktipis

et al.'s (2018) broader concept fundamental to understanding human cooperative relationships, fitness interdependence (see also Aktipis et al., 2011; Aktipis et al., 2016; Cronk et al., 2019). And the kernel of Tooby and Cosmides's insights have been broadened and formalized by the concept of "stake-based" cooperation and relationship investment (e.g., Roberts, 2005; Barclay & Van Vugt, 2015; Barclay, 2020; Barclay et al., 2021; Pleasant & Barclay, 2021). The functions of friendships have typically discussed them as dyadic relationships. Yet, as just noted, people typically have multiple friends. And friends of a target person may have relationships with one another; people have friendship networks (e.g., Vieth et al., 2025). As we develop below, key ideas about the function of friendships imply that each friendship a person has may affect the strength, nature, and benefits derived from the person's other friendships. In particular, the addition of friendships may potentially weaken the benefits derived from other friendships, though a person's management of their multiple friendships may mitigate these impacts. Hence, people should not only possess adaptations for developing and maintaining dyadic friendships; they should have adaptations for developing, maintaining, and regulating *friendship networks*.

Our paper is conceptual in nature. First, we provide an overview of key ideas about the function of friendships and how these relationships are distinct, focusing on the status as stake-based relationships (Sections 2-4 below). We observe that an implication of these ideas is that a person's friendships may impact the potential benefits from other friendships. Second, then, we discuss the implications of these ideas for understanding adaptive problems associated with managing friendship networks and their possible solutions (Sections 5-6). Third, we discuss key domains for future conceptual and empirical development, as well as potential applications (Section 7).

2. Fitness Interdependence

Individuals within a population may influence the inclusive fitness of one another, the replication of their genes. The degree to which they do so reflects *fitness interdependence* (Aktipis et al., 2018). Competition between individuals generates negative fitness interdependence: To the extent that one individual succeeds in a competition (e.g., for resources, territories, or mates), another individual (or set of individuals) often loses. Cooperation generates positive fitness interdependence. Even when two conspecifics pursue their own fitness interests, they may positively affect each other's fitness if, in seeking a common goal, they aid the other in achieving the goal as well (e.g., when two individuals cooperatively hunt; Aktipis, 2004; Noë & Hammerstein, 1994; e.g., in hyenas, Holekamp et al., 2007; Smith et al., 2012). Positive fitness interdependence, then, is common in animals, including, notably, nonhuman primates (e.g., Dunbar, 2018).

Humans, however, appear to be unusual in the degree to which non-kin, non-mating relationships exhibit positive fitness interdependence (Wrangham & Benenson, 2017). Not only do people extensively engage in mutualistic cooperation (e.g., in hunter-gatherer groups, cooperative hunting). Through reciprocal altruism, two individuals iteratively provide aid to each other, especially on occasions when the benefit of aid (in the currency of fitness) to the recipient exceeds the cost to the giver (Trivers, 1971). Sharing of food surpluses, then, is extensive in human groups (e.g., Gurven, 2004). Similarly, individuals may cooperate to care for each other's dependent children, while managing other tasks, such as food-gathering and preparation (e.g., Hrdy, 2009). Recently, Balliet et al. (2016) conceptualized taxonomies of situations of interdependence introduced in the social psychology literature by Kelley and Thibaut (1978) as recurrent adaptive problems for which people have evolved adaptations (Balliet et al., 2016).

People distinguish their closest friendships from other forms of social relationships, even those in which extensive cooperative activity occurs. Tooby and Cosmides (1996) sought to

understand the evolutionary scenario that could have shaped adaptations for people's propensities to form the kind of close relationships people identify as friendships.

3. Deep Engagement and Stake-Based Interpersonal Investment

In pure exchange relationships, individuals expect an immediate return to any benefit offered to a partner (e.g., in the context of direct barter). In a relationship characterized by reciprocal altruism, individuals tolerate time lags between exchange of benefits. Naturally, if only one individual offers benefits to the other, altruism is not reciprocal, and the relationship will cease to exist. Still, participants need not expect immediate return to investment. In communal relationships, individuals provide benefits to the other, often based on perceived need of aid, with no expectation of benefits in return (Clark & Mills, 2012). Close friendships are construed as one important form of communal relationship. The high levels of fitness interdependence in human friendship are not sufficiently explained by mutualism or reciprocal altruism. Tooby and Cosmides (1996) proposed a distinctive explanation of the "deep engagement" that characterizes human friendship.

Suppose two individuals develop a relationship through which the actions of each individual have positive fitness consequences for the other, whether through mutualistically beneficial activities or reciprocal exchange. Over time, individuals come to know each other's capacities and needs, which facilitate the coordination of mutually beneficial cooperative activities. And they come to know them in a way that they do not know others' capacities and needs. This knowledge of each other's features renders the other a favorable cooperative partner, potentially across many different contexts in which cooperation could be mutually beneficial. As an outcome of a history of repeated assistance and mutualistic collaborations ("deep engagement;" Tooby & Cosmides, 1996), they become highly valued social partners for each other, ones who are not readily replaced.

The very substantial bidirectional positive fitness interdependence of these kinds of relationships has an important set of consequences that further deepen them. As each partner is a

highly valued form of fitness-enhancing social capital to the other, each partner is motivated to protect that social capital. Any threat to one partner's well-being is not only a threat to that individual's fitness; it is simultaneously a threat to the interdependent partner's fitness. The benefits of this sort of social relationship, then, go well beyond the benefits of day-to-day, "normal," reciprocation-based cooperative activity.

Tooby and Cosmides (1996) presented a verbal argument regarding the benefits of cooperation characteristic of friendships, distinct from reciprocity alone. More recently, theorists have developed formal models that capture Tooby and Cosmides's arguments while also broadening them. Though the basic concept underlying the shared conceptualization goes by different names (e.g., partnership: Eshel & Shaked, 2001; byproduct reciprocity: Sachs et al., 2004; group augmentation: Kokko et al., 2001; for a related concept, see Connor, 1986, on pseudo-reciprocity), recent scholars have used the term "stake" (Roberts, 2005; see also Barclay, 2020, Barclay et al., 2001; Pleasant & Barclay, 2024). We, therefore, will refer to "stake-based" cooperation or interpersonal investment.

In a simple cost-benefit model, Barclay (2020) showed how a relationship based on reciprocation of benefits can transform into one involving stake (in addition to reciprocation and mutualism). Suppose two individuals exchange reciprocal benefits, where, following Trivers (1971), the benefit to one partner (b) exceeds the cost paid by the other partner (c), such that, on each "round" of exchange for each partner, $b - c > 0$. If, over time, the expected (even if uncertain) number of future rounds is n , then the total future benefit of the cooperative relationship is $(b - c) \times n$. If, at the end of each round, there is a risk that a partner will die or be unable to provide future benefits, it will benefit an individual to pay cost a to "save" the partner, so long as $a < (b - c)n$.

The benefit of each individual of helping a partner in these cases is not based on direct reciprocation; it is based on the self-benefiting stake the individual has in continuing the

relationship. This process has two important implications. First, it offers a basis for understanding what leads to and maintains need-based delivery of benefits: If one function of benefit-delivery is to maintain the well-being of a valued interaction partner, it makes sense that benefit-delivery is sensitive to the partner's needs. Second, it has implications for the conditions of benefit-delivery. Reciprocation-based delivery of benefits entails knowledge of the other who delivered benefits. Yet, stake—the benefit of each partner to the other's well-being—can maintain *anonymous* delivery of benefits (Barclay, 2020; Pleasant & Barclay, 2024). In short, stake-based investment offers a functional explanation of how an exchange relationship may be transformed into a communal relationship (e.g., Clark & Mills, 2012) —ones we refer to as “true” friendships.

Barclay (2020) also showed that replaceability of a partner affects stake-based investment in them. When an individual can readily find an alternative partner who can deliver similar future benefits through reciprocation that the current partner does, the individual's stake in the well-being of the current partner is reduced, meaning that (1) the threshold cost, a , that it pays to “save” the current partner is reduced, and (2) each individual's willingness to anonymously deliver benefits to the current partner is also reduced. Tooby and Cosmides (1996) emphasized that friends become irreplaceable because of the history of the ability to reciprocate aid to the other is not immediately available with an alternative partner. This process may be partly accounted for by incremental increases in willingness to cooperate over time—“raising the stakes”—, which have been modeled empirically by Roberts and Sherratt (1998) and Roberts and Renwick (2003). This involves another process—the efficiency of coordination of mutual benefit-delivery owing to partners knowing each other—, which increases the $b - c$ difference in reciprocation as well as the effectiveness by which cooperative mutualism delivers benefits to each party. However, any reason for relative irreplaceability (e.g., an individual does not have enough competence to be an attractive partner to other potential partners) can increase an individual's stake in their current partner. This raises an

interesting dynamic: Highly competent and valuable partners should have greater partner choice. As a result, they may have less stake in the well-being of their current partners. If, however, their partners perceive that their relationships are less stable, partners may come to have less stake in relationships with highly competent individuals, all else equal. Of course, all else is not equal because highly competent individuals also offer benefits. Less stability due to partner choice, therefore, trades off against those benefits. Depending on their relative impact, an individual could, in theory, obtain greater benefits from a less competent but more stable partner than a more competent but less stable partner due to the deeper engagement and communal character of the former relationship.¹

As we develop further below, the observation that irreplaceability of a partner increases stake-based investment in the relationship, whereas replaceability of a partner diminishes it, has important implications for our core concern in this paper, the impact of a person's friendships on the nature of the person's other friendships.

4. Crisis Management

Tooby and Cosmides (1996) emphasized the adaptive value of friends in *crises*—when an individual's viability is threatened to the point that they could die or be severely harmed without assistance. In their words, “For hunter-gatherers, illness, injury, bad luck in foraging, or the inability to resist an attack by social antagonists would all have been frequent reversals of fortune with a major selective impact” (p. 132). Furthermore, “The ability to attract assistance during such threatening reversals of welfare, where the absence of help might well be deadly, may well have had

¹ Barclay (2020; see also Barclay et al., 2021) noted that stake-based friendships involve fitness interdependence but also clarified this connection. When individuals have a stake in the well-being of another individual, they benefit from the other surviving and remaining healthy. That is, they benefit from the fitness of the other. Yet, they do not necessarily need to benefit from the *reproduction* of the other. (Indeed, an individual may constitute a better friend if they don't have kin and offspring to invest in.) Fitness interdependence, then, may be a broader concept than what is pertinent—interdependence centers around survival, health, and well-being.

far more significant selective consequences than the ability to cultivate social exchange relationships that promote marginal increases in return during times when one is healthy, safe, and well-fed” (p. 132). An individual’s own “extinction” has powerful fitness consequences, not only regarding their own inability to reproduce, but also via the viability of their kin and offspring (e.g., Ellison, 1994; Hill & Hurtado, 1996).

Friends often engage in exchange and mutualistic collaboration during good times—indeed, it is these kinds of benefits that lead friends to become valued social capital. Tooby and Cosmides (1996) implied, however, that the greatest fitness benefits that shaped psychological designs pertaining to true friendships may have centered on friends’ willingness to help each other during crises, which occurs once friends become highly valued social capital to each other. As implied above, friends may be willing to incur substantial costs to help each other because the future returns—in reciprocal exchanges of benefits, $(b - c)n$ —may be substantial, due both to large $b - c$ and large future anticipated n . Friends, therefore, serve a function similar to insurance policies against the need for medical assistance or property loss: During times of crisis, each delivers a big benefit that offsets potential catastrophe. One key difference, of course, is that whereas insurance companies pay benefits that are contractually owed, friends deliver benefits because *it pays them* to do so. Nonetheless, the basic notion of insurance is applicable. One invests in an insurance policy precisely when the future (under uncertainty) could potentially impose a massive cost (e.g., loss of valuable personal property, a medical crisis) that one cannot immediately offset with their personal assets. Thus, investment in insurance is a cost paid presently to hedge against the massive cost of a potential future crisis. Investment in friendships, in Tooby and Cosmides (1996) conceptualization, is akin to investment in insurance against catastrophic loss.

The idea that communal relationships offer protection (risk-reduction) in the face of crises is explicit and culturally crystallized in the East African Maasai notion of *osotua* (“umbilical cord”)

relationships: Lifelong friendships in which gifts, such as livestock, are given when crises to the other party occur, with no expectation of return (e.g., Aktipis et al., 2011, 2016; Cronk et al., 2019).

4.1 Risks during Crisis Management

In Tooby and Cosmides' (1996) view, deep fitness interdependence between friends hinges on their mutual irreplaceability. Each friend's fitness depends in important ways on the benefits delivered (or that can be delivered) by the other because the same benefits cannot be readily delivered by other partners. Moreover, if an individual lost the friend, it would be difficult to develop another relationship that would offer similar benefits. Precisely because friends depend on each other for important fitness benefits, each becomes committed to the other and invested in protecting both their friend's as well as their own social capital. Furthermore, to the extent that each friend knows the other's level of self-interested commitment to the relationship, each partner is more likely to trust the other to deliver benefits when needed. Indeed, friends are more likely to be tolerant to imbalances when one friend is needs assistant (compared to strangers), however, friends still monitor or track one anothers self-interested commitment to garner information regarding their friends selflessness towards them (Xue & Silk, 2012).

If a friend becomes a unique and indispensable social resource, one can imagine an idealized world in which very close friends pair up, two-by-two, each being the other's one true, irreplaceable friend and distinctly trustworthy insurance against catastrophe. (Indeed, this view matches well with Aristotle's ideal conceptualization of a friendship of virtue, where a friendship entails "a single soul dwelling in two bodies" [Diogenes Laertius, 2018, Book V, Section 21].) This idealized world, of course, does not reflect the real world. Although people often consider someone to be a "best" friend, they typically have and seek to develop multiple close friendships. Upon reflection, there is a good reason there might be benefits to having multiple friendships.

4.2 Risky insurance reliability

Even though one close friend may be highly committed to help in times of need, they may not always be able to deliver help. For example, the friend may die or become incapacitated. Perhaps more frequently, friendship partners' times of need may coincide: Just when one partner has a run of bad luck in foraging, the other suffers an injury. Having two or more friends-as-insurers hedges against risks that one of them will not be available at a future time of need. It also hedges against the risk that a person overestimates the extent to which a partner has stake in their relationship (see Clark & Mills, 2012, on the impacts of overestimating a partner's willingness to invest in a communal relationship).

4.3 Portfolio Diversification

Though Tooby and Cosmides (1996) stressed the fitness benefits of friends in times of welfare reversals, they also recognized the value of friendship partners in reciprocal exchanges and mutually beneficial cooperative pursuits, even during good times. In these contexts, however, a singular friend may be better suited to some cooperative activities rather than others. For instance, a friend may be a superb animal tracker, but not physically strong. Having multiple friends, each with a different set of competencies, can not only diversify an individual's insurance against different types of welfare reversals; it can also diversify the kinds of mutually profitable cooperative endeavors that multiple friends make possible. Recent work on the distinctions between local in-group friendships and long-distance friends provide evidence for the benefits of diversification across friend networks: While local and in-group friendships benefit group cohesion and the minimization of conflicts, long distance friends may be relied on for support during resource shortfalls (Pisor & Ross, 2022).

4.4 The need for numbers

The adequacy of a singular friend-as-insurer hinges on their ability to deliver needed benefits during dire circumstances. In some circumstances, however, a single friend may be unable to deliver

the benefits needed. If, for example, a parent becomes ill and is unable to forage and keep their family fed, a singular friend may not be able to generate enough food to cover these deficits at reasonably low costs. By contrast, multiple friends may each be able to generate small surpluses at reasonably low costs to themselves such that, in aggregate, the full deficit gets covered. Tooby and Cosmides (1996) discussed one dire circumstance that may demand numbers: “the inability to resist an attack by social antagonists” (p. 132). If the attacking social antagonists number three, the willingness of a single friend to join the resistance may result in the demise of both partners. But if an individual can call on four friends to help defend, resistance may succeed.

This brings us to an alternative view of the benefits of friendships: Friends may be alliance partners rather than exchange partners (e.g., DeScioli & Kurzban, 2009). Alliance partners are valuable precisely because of the benefits they can offer during social conflicts with others. To the extent that individuals are part of large and powerful alliances, they are likely to have access to a network of social resources that can be brought to bear when conflicts with individuals outside of the alliance arise. Powerful alliance partners are those who also have a large number of alliance partners. According to this view, single friends only incrementally add value; ultimately, individuals benefit more from cultivating friendship networks, which allow them to attract new alliance partners in addition to those they already have. Selection pressures for social alliances may be shared with chimpanzee males, where despite considerable within-group violence, male chimpanzees form coalitions to patrol the border of their territory (Wilson et al., 2014; Wilson & Glowacki, 2017). In addition, this perspective is not necessarily independent of stake-based investment in friendships given that friends can both serve as members of an alliance *and* be social capital worth investing in.

5. Special Adaptive Problems of Maintaining a Friendship Network

Tooby and Cosmides (1996) sought to identify and delineate adaptive design for establishing and maintaining friendships—both their nature and what adaptive design could reveal about the

selective forces that shaped them. In so doing, they focused on dyadic relationships. Modeling of stake-based relationship investment has similarly focused on dyadic relationships (e.g., Barclay, 2020). Individuals' multiple friendships constitute friendship networks. To the extent that special adaptive problems arose ancestrally with regard to establishing and maintaining friendship networks, we might expect that people possess evidence of adaptive design for establishing, maintaining, and regulating friendship networks, where a network involves "true" friendship with multiple others. We next discuss some of the special problems that arise in this context. While we offer no explicit modeling of expected outcomes, we hope to stimulate interest in understanding the dynamics that occur when dyadic friendships exist in the context of broader networks, discussing some of the key parameters that should be considered.

We focus on one fundamental issue that has multiple manifestations and no obvious single solution. *As the number of stake-based friends increases, the stake one has in each friendship tends to decrease.* We previously discussed the condition that leads to this expectation: Irreplaceability fosters stake. Replaceability diminishes stake. Addition of a friend may well have marginal value. But, at least in part because of replaceability, all else equal, the marginal value is reduced. What's more, because addition of the friend renders other friends less irreplaceable as well, addition of a friend diminishes the marginal value of all other friends as well; everyone becomes less irreplaceable. Friends, of course, are agents with their own self-interest. *If friends in a network perceive that one's own stake in a friendship diminishes with network size, friends may diminish their own investment in the relationship, as their trust in the relationship is expected to weaken.* That is, as one's own stake in a friendship is reduced, the friend's reciprocal stake in the friendship diminishes in turn. And, as the number of stake-based friendships increases, this dilution effect may spread across one's full network.

In the extreme, these impacts are captured by a quote famously attributed to Aristotle: "He who has many friends has no friends" (Diogenes Laertius, 2018, Book V, Section 21). This

fundamental issue associated with friendship networks, which cannot be understood by models of dyadic relationships alone, has several implications for the adaptive problems individuals face with respect to managing their friendship networks.

5.1 *The problem of number*

Just as one must consider cost-benefit trade-offs when deciding whether it is worth it to back up an unreliable insurance policy with another policy, adaptive design demands solutions to trade-off problems when deciding whether to develop new true friendships. Additional friendships bring benefits, but their development and maintenance also entail costs and commitment, as friends “bank” on their partners to be friends-as-insurers. These commitment costs have been well-recognized (e.g., Dunbar, 2018). In addition, however, we add additional considerations: The impact of increased numbers of friends on the stake one has in each friendship and, accordingly, the stake each friend has in oneself. Adaptive design for friendship networks should include ways of assessing the relative costs and benefits of investing in additional friendships.

5.2 *The problem of commitment and trust*

Related to the issue of friendship number, but partly separate as an adaptive problem, is the issue of commitment and trust. When two people function as each other’s only true friend, each partner affects the fitness of the other substantially. Each is a highly valuable, difficult-to-replace form of social capital for the other. As a result, it is in the interest of each friend to protect that social capital, that is, to supply aid when the other faces major threats or challenges. Each friend, therefore, becomes committed to protecting the welfare of the other. As we have emphasized, however, additional friends may dilute irreplaceability and, accordingly, stake. Again, not only does the new friend potentially add less value to the benefits that friends who are already in the network provide; the addition of a new friend may, ironically, *subtract* from the value of other friends already

in the network. Cast another way, there are risks that each friend becomes, in a sense, less of a true friend.

Are there ways individuals can bolster commitment and trust within friendships in the context of friendship networks, thereby mitigating the potential costs of adding new friends? We consider possible adaptive solutions below. Here, we note that the task of maintaining the value of individual friendships within friendship networks constitutes an important adaptive problem that becomes pertinent to understanding psychological design for friendships primarily within the context of larger friendship networks.

5.3 The problem of coordinating friends and managing conflicts of interest within friendship networks.

Suppose an individual experiences a serious threat to their well-being (e.g., an illness, an injury, a run of bad luck foraging). When one has a singular friend-as-insurer, who one can and should count on to help is clear: the singular friend. But when one has multiple friends, the responsibility is less clear-cut; the distribution of costs incurred and benefits reaped by friends become more complicated. Each friend, for example, values their partner, the reason it pays them to provide aid. But for that reason, each friend also benefits from aid given by other friends: When Friend 1 helps the individual, Friend 2 benefits too, even if Friend 1 has no interest in Friend 2's own welfare. As both friends benefit, but only one pays costs, conflicts of interest between friends can emerge (see, e.g., Olivola et al., 2019, on the “volunteer’s dilemma”).

Matters may be more complicated, as who delivers aid may garner greater future benefits if the recipient experiences greater stake in the relationship with the direct giver. Each friend, then, may want to be “credited” with giving aid in the future, motivating one or both friends to “compete” to offer aid (e.g., Barclay et al., 2021; see also Barclay, 2011, 2013, 2016, on forms of competition and signaling within “biological markets”). Moreover, each friend may benefit from

being seen by the individual as having offered more aid than they truly delivered, producing additional conflict between friends (see Krems et al., 2021, 2022,).

Sustained conflicts of interest between friends should also introduce inefficiencies within an individual's network of friends. In an efficient system, all of friends' efforts to deliver aid directly benefit the individual in some way. However, when conflicts of interest between friends exist, some effort exerted by each friend may go toward increasing their own net benefits, independent of whether or how the individual is benefited. For instance, when friends compete to be the one responsible for giving aid, each friend could at times benefit from interfering with the other friend's efforts to give aid (e.g., Krems et al., 2021, 2022, 2023). Such conflicts of interest impose costs on each friend within a network, which usually do not translate into benefits for the individual. They may furthermore impose costs on friends in a network, possibly leading them to leave it (i.e., reduce stake in the friendship).

One should expect, then, that individuals possess adaptive design for managing the conflicts of interest within their friendship networks, both conflicts of interest between friends but also conflicts of interest they have with friends of other friends. Krems and colleagues' research has explored the nature and implications of the conflicts of interest that can arise within networks of friends, including the nature of psychological design needed to manage these conflicts of interest (e.g., Krems et al., 2021, 2022, 2023; see also Ayers et al., 2024; Merrie et al., 2024; Zankich et al., 2025). Relatedly, Krems and Conroy-Beam (2020) have documented that friendships emerge within social markets that in some ways are similar to how mateships emerge within mating markets. That is, just as people not only choose mates but also compete with others to be chosen as mates, people not only choose friends but also compete with others to be chosen as friends. Finally, people may be wary of too much coordination between friends as it is likely to suggest that Friend 1 & Friend 2 occasionally talk about Friend A. While this has many potential benefits, it also allows for possibility

that Friend 1 and Friend 2 develop a closer friendship and become less invested in the original focal friend. Because of this, people are likely to be vigilant of their friends interactions with each other and monitor the closeness between them.

6. Potential Solutions to the Adaptive Problems of Maintaining Friendship Networks

We now turn to ways in which people may address the special adaptive problems associated with developing, maintaining, and managing friendship networks. Our proposals are provisional. In some cases, which solutions are most adaptive may depend on the relative costs and benefits that are unknown or may vary across contexts. In what follows, we lay out key domains that adaptive design should be expected to address.

We discussed three potential adaptive problems. First, there's the issue of how many true friendships should an individual invest in. Second, there's the question of how to build and maintain commitment and trust within a friendship network. Third, there's the issue of how to regulate and manage potential conflicts of interest within a friendship network.

6.1 *Diversification within networks*

As mentioned above, one benefit of having multiple friends is that adding friends can broaden the competencies and skill sets represented within a friendship network. People who foster networks of friends that fulfill different emotional needs and thus, allow them to diversify those needs across the network, have better well-being (Cheung et al., 2014). It makes sense, then, that the marginal value of an additional friend depends not only on the potential new friend's competences, but also on the competencies already represented within an individual's existing friendship network. Thus, adaptive design for friendship networks should include solutions that assess and evaluate the marginal value of friends in this regard. In one study, people were more likely to value individual traits rather than group membership when selecting a partner for cooperation (Pisor & Gurven, 2018). suggesting that people may seek out new friends with specialized individual traits.

Diversifying friends' talents and competencies may also partly address issues pertaining to commitment and trust. A friend who has competencies that uniquely deliver benefits, despite being one of multiple friends, remains, in this sense, irreplaceable. One hence retains greater stake in such a friendship. And, to the extent that the friend recognizes the distinctive benefits they offer, the friend too has reason to retain commitment to and trust in the friendship. All else equal, then, specialization within a network of friends should stabilize most networks. Individuals may also benefit from developing and making known their specialized talents within a friendship network, which might also make them irreplaceable to other network members. Finally, there is emerging evidence that engaging with weak ties (casual connections within one's social network that are not defined by a relationship title) within one's network has benefits for well-being (Sandstrom & Dunn, 2014). Maintaining weak ties may be a way to diversify one's network without investing the substantial, and sometimes risky, investment into a singular or set of true friends.

6.2 Potentials solutions to both increase commitment to and ameliorate conflicts of interest within friendship networks: Building networks of friends

As discussed earlier, conflicts of interest between friends can generate inefficiencies in the friendship network's delivery of benefits. By imposing such costs, they may reduce the benefits that are delivered to an individual. They may also undermine friends' commitment to friendships within the network. So how might individuals reduce and manage conflicts of interest within their friendship networks? One way to do so seems straightforward: Promote friends' shared interests (i.e., the extent to which they benefit from similar positive outcomes). Friends within a network may often have shared interests if *they* are true friends—that is, when they share histories that have led them to experience deep engagement and strong commitment to one another's well-being. Indeed, modeling work shows that choosing friends who are similar to oneself leads to having friends who are similar to each other, which can be beneficial by facilitating friendships between friends (David-

Barrett, 2020). Furthermore, one potential solution rests in one's centrality within their network of friends.

In our discussion above, we considered how two friends of an individual might respond to the individual's current need. Conflicts of interest can arise for several reasons (e.g., if Friend 1 helps the individual, Friend 2 should benefit some because Friend 2 also values the individual, meaning that Friend 2 receives a benefit that Friend 1 provided). Now suppose that Friend 1 and Friend 2 are also true friends who care about each other's outcomes, not just the individual's outcomes. If so, when Friend 1 helps the individual, this benefits Friend 1 in two ways. First, it helps the individual while also bolstering Friend 2's welfare, which Friend 1 values as well. Second, the cost that Friend 1 pays to aid the individual is also a cost to Friend 2, given that Friend 2 values Friend 1's outcomes. Thus, if all three individuals are true friends with each other, conflicts of interest over helping each other are reduced, relative to a network in which an individual has two friends who are not friends with each other. Earlier, we also considered the possibility that Friend 1 and Friend 2 might compete to offer aid to the individual to receive "credit" for giving aid. Once again, conflicts of interest between Friend 1 and Friend 2 should diminish if they are friends and care about each other's well-being (David-Barrett, 2020).

We can extrapolate from a network of three friends to networks of four, five, or n friends: Individuals may benefit from being part of friendship networks in which everyone within the network are mutual friends—that is, from not just having a "friendship network" to being part of a "network of friends." This network structure may enhance each friend's commitment to each other's friends. And there is another benefit as well. Earlier, we noted that an individual in dire need may be able to obtain better aid from multiple friends at lower cost to each friend. Individuals may particularly be able to do so effectively and efficiently when the multiple friends are friends themselves, part of the same "team" of individuals with stronger shared interests.

If people benefit from being part of a network of friends, individuals should attempt to solidify their network by encouraging friendships between their friends. Hence, they may work to bolster the fitness interdependences that exists between their friends. Individuals may also see value in fostering relationships with people who are friends of their most highly valued friends because doing so may increase their original friend's commitment to the friendship.

6.3 Implications for network structure

Increasing ties within a network has implications for network structure. Though networks can be distinguished in a host of ways (e.g., Oldham et al., 2019), one key quality is the extent to which a network is hub-dominated, with one or very few people occupying “central” positions with relatively many connections to others and others with few connections to others (high centrality variance), vs. a network that is egalitarian, with most people in the network with connections with others (low centrality variance). When a friendship is a network of friends, as we describe above, the network has low centrality variance. Even in rock hyrax, low centrality variance in social networks was found to be associated with survival within the group (Barocas et al., 2011). In human friendship networks, low centrality variance may be associated with greater network stability and efficiency with which benefits are delivered to others. By contrast, a classic, hub-dominated “star” network, with one central individual connected to, say, five others, none of whom have connections with any others, fosters potential conflicts of interests between all five peripheral persons. [l.]

6.4 The risks associated with a network of friends

The value of having a network of friends notwithstanding, developing and maintaining such a network poses considerable challenges. In an idealized world, each pair-wise friendship within a network would be equally and strongly valued by all partners. In such an idealized world, each friend within the network would be equally committed to the entire network or “team.” Such an idealized situation, however, is difficult to achieve and maintain.

Within a network, each individual has a set of talents that can, in times of need, benefit others within the network. Each person also has a set of vulnerabilities to, at some point, be in a state of need requiring aid from others. Given these individuating features, not every person within a network of friends is likely to be equally valued by every other person. This should be particularly true as the size of a network increases and the number of pair-wise relationships increases exponentially. For instance, doubling a network of friends from 3 to 6 increases the number of pair-wise relationships within the network by five-fold, from 3 to 15. At the same time, once again, each friend's irreplaceability may diminish with increased network size. As Krems and colleagues have shown (e.g., Krems et al., 2021, 2022), people may be motivated to increase their irreplaceability to their friends by, for example, becoming a "best friend" relative to a friend's other friends. When inequalities in strength of bonds between friends exist within a network, conflicts of interest between friends may also arise. Dunbar (2018) has observed that people tend to have roughly five close, intimate relationships, in large part because of the considerable time and psychological costs of developing and maintaining friendships. Here, we observe another consideration: As network size increases, so too does the potential for inefficiencies in benefit-delivery in friendship.

As we've noted, when such conflicts exist, they can threaten the stability of a network. Pair-wise conflicts of interest may lead some within the network to prefer to be part of a smaller network, excluding those with whom conflicts are greatest. Larger networks of friends, then, have the potential to splinter into smaller, more stable networks. That said, some individuals may benefit more from friendships within two different networks, where people within each group are not friends with those in the other group. Effective management of friendship networks, in other words, does not always involve forming one network of friends. Often, true friendships might be more stable and offer greater net benefits when an individual's friends are *not* all true friends with one

another. Effective management, therefore, requires understanding the nature of the relationships that exist (and are possible) between friends within a network.

6.5 Information-seeking within friendship networks

As Tooby and Cosmides (1996) recognized, friendships are founded on histories that promote and ensure mutual trust—trust that partners will offer aid in times of need. Mutual trust, in turn, is partly founded on mutual understanding—each partner’s understanding of the other’s needs and how their own capacities enable them to satisfy those needs. Effective management of a friendship network entails mutual understanding of this sort. However, it also entails understanding the relationships that exist between different pairs of friends within a network (or smaller sub-networks). Are one’s friends also true friends to each other? If so, what kind of mutual understanding maintains that trust? If not, what are their potential conflicts of interest, and how might those conflicts affect each friend’s costs and benefits of maintaining a friendship with oneself? How can potential conflicts of interest between friends be mitigated or entirely avoided? How might relationships between certain friends introduce conflicts of interest between oneself and other friends? Could one friend seek to compete with oneself to become a better, more irreplaceable partner to another of one’s friends? As an individual’s friendship network increases in size, so do the informational demands of managing it effectively, which may have many limitations as they are cognitively costly (Dunbar, 2009). For this reason, people attend to and track this type of information about their friendship networks.

6.6 When numbers matter

We previously noted the benefits of having multiple friends, particularly when aid requires assistance from several others (e.g., when friendships are especially beneficial by serving as alliance partners). A prime example is when social antagonists threaten an individual, who may benefit the most when a small “army” of defenders respond (or when social antagonists understand that a small

army of defenders is available to respond). In this case, strength of aid scales with the number of responders, which may be one exception to the fundamental premise that the addition of friends typically weakens the quality and net benefits of existing friendships. When friendship networks offer benefits to constituents that scale with the number offering aid in this manner, one should anticipate greater network stability on average. Each friend within the network provides an incremental benefit to all others within the network. This feature of friendship networks may be especially important for individuals who have specific needs or are at particular life stages, as discussed in greater detail below.

6.6 Summary

People not only have adaptations for developing, regulating, and maintaining the special class of close, high fitness interdependence relationships referred to as friendships; they should also have adaptations for developing, regulating, and maintaining *networks* of friendships. We identified several specific adaptive problems that arise in the context of friendship networks. And we offered provisional thoughts on the types of solutions that individuals possess in response to these adaptive problems. Some important exceptions notwithstanding, these adaptive problems and their solutions have not received much attention from an adaptationist perspective. At a global level, then, our discussion of these issues aims to encourage additional conceptual and empirical work to address them. We now turn, however, to suggest directions for future progress.

7. Future Directions and Potential Programs of Research

7.1 Evolutionary Modeling

While the model we present is primarily verbal, we aim to stimulate the development of formal evolutionary models capable of capturing the complexity of human friendship networks. Formal models and simulations of the evolution of cooperation have a long, productive history in evolutionary biology (Maynard Smith, 1984; McElreath & Boyd, 2007; Roberts, 2005). In contrast,

psychological theories often fail to generate a rich set of precise predictions that decisively test alternatives (Meehl, 1990). Echoing recent calls to prioritize stronger theoretical development over incremental hypothesis testing (Scheel et al., 2021), we suggest that the kinds of questions we raise about how people effectively manage friendship networks can be addressed with simple but generative formalizations of the underlying adaptive problems, where existing dyadic models (e.g., Barclay et al., 2021) are modified to incorporate features of broader social networks.

A useful example exists: Leimar and Bshary (2024) developed a model of the emergence and maintenance of social bonds between unrelated group mates, inspired by blood (food) sharing in vampire bats. In the model, individual “bats” learn who has shared blood with them, which accordingly increases their bondedness to (and fitness interdependence with) particular individuals. Bats then share increasingly with these individuals, along lines of stake-based cooperation. Consistent with “social bet-hedging” (Carter et al., 2017), bats remain open to developing social bonds with new partners, but optimally do so only by not sacrificing strong bonds with established partners. Key parameters, such as rate of developing social bonds, were allowed to evolve in the model; especially when individuals could identify helping partners and could engage in partner choice, parameters did evolve in ways that support stake-based cooperation (see also Carter, 2024).

This model may be profitably edited and extended to incorporate additional elements of human friendship networks, along lines we discuss. Leimar and Bshary (2024) modeled large groups of individuals, but they only modeled multiple dyadic relations. One next step would be to extend this model to include instances where exchanges between two individuals impacts the quality of other relationships. Does competition between individuals to more strongly bond with particular individuals emerge? Does bondedness between friends of friends (emergence of “networks of friends”) reduce conflicts of interest? How, in particularized ways, do individuals trade-off deep engage with particular, “irreplaceable” partners, and establishment of broader networks, and what

parameters influence this trade-off? (E.g., overall competence of individuals may affect their opportunities to form new bonds.) How do overall mortality/morbidity rates, and the impact of patterns of helping on these rates, affect the dynamics of friendship network formation and maintenance? How do dynamics change when the benefits of larger alliances are built into the models? When strength of associations between kin are added into the model, how is friendship formation affected? Outcomes could importantly inspire function-form predictions that permit “riskier” hypothesis testing (Meehl, 1990).

7.2. New Directions for Empirical Research

The perspective we lay out proposes that adaptations for friendship formation and maintenance are embedded within larger sets of adaptations that function to develop, maintain, and regulate friendship networks. To assess how such adaptations operate, research should examine how these processes dynamically unfold over time, and how those dynamics depend upon network features, such as variability in centrality and commonality of interests across networks. As multiple relationships must be tracked in this research, optimally viewed from the perspective of all individuals involved, this work is challenging to conduct but, ultimately, necessary to inform our understanding of friendship networks.

7.3. Lifespan Perspectives and Gender

The types of friendship networks people cultivate (e.g., their size, diversity, cohesion) may vary depending on challenges people currently face and crises that might arise. Specific challenges and threats to survival and reproduction may shift depending on stage of the lifespan and gender. Thus, people may find that maintaining and managing friendship networks require “network tailoring” over time based on their current life stage and goals. Below, we discuss a few possible ways in which network management may vary across lifespan or genders, with an aim toward stimulating research questions.

In adolescence and young adulthood, securing friends and gaining social status is a key developmental challenge for both men and women (Hartup & Stevens, 1999), partly associated with the task of finding mates. Having a large friend network may foster social potency and upward mobility (Ball & Newman, 2013). As well, friends may provide valuable information regarding potential mates, help to facilitate pairings, or increase either one's own or a potential mate's perceived value (Ackerman & Kenrick, 2009; Russell et al., 2013). When friends vouch for or provide information on potential mates, it may also make the choice of pursuing them feel safer or less risky. Thus, these life stages and their associated goals may favor broader and more diverse friendship networks (Hartup & Stevens, 1999), while at the same time demanding special care to regulate friendship networks.

Relatedly, at these life stages, substantial, cohesive networks may especially function as alliances that protect individuals during conflicts (DeScioli & Kurzban, 2009)—perhaps particularly for men, as they more likely engage in physical conflicts, whether they be intra-group or inter-group (Archer, 2009; McDonald et al., 2012). Once again, then, regulation of broad friendship networks may be a key life task during these stages.

Parenthood brings its own challenges to maintaining social networks. Task-specific sex roles may vary considerably across foraging societies (e.g., as a function of reliance on large game vs. small game in the diet, proportion of meat in the diet; e.g., Marlowe, 2007). But, in broad strokes, male investment in offspring tends to focus more on provisioning; in some societies, women forage more of the calories but still are typically more involved in direct childcare (e.g., Marlowe, 2007; Kaplan et al., 2000). Men and women, then, may have evolved sex-differentiated special designs to choosing friendship networks that provided solutions to these problems (see Taylor et al., 2000). Because large game hunting usually consists of many males working together in a cohesive group, followed by widespread sharing of the calories obtained (e.g., Dyble et al., 2016), having wider friendship

networks with flexible ties may have been beneficial, though notable exceptions are possible (e.g., having one very close friend with complimentary hunting skills). In contrast, for child-rearing, women may have benefited from relying on a few close, highly trusted friends, alongside kin. Focal, trusted friendships may have been especially important for women in groups lacking close kin. One recent study found that women, more than men, said they had a best friend, though also reported more friendships (Dunbar et al., 2024). Men, more than women, reported that their friends were interconnected. Again, adaptive problems associated with network regulation may be sex-differentiated.

Later in life, as physical mobility wanes, a smaller but supportive network of friends may become important. For women, friends may shrink in number as kin take on more duties and grandmothers invest in grandchildren (Hawkes & Coxworth, 2013). Though exceptions (e.g., men with high status) exist, most older men do not effectively compete for new mates; they may benefit from investing in children and grandchildren too (Gangestad & Simpson, 2000). On average, close, trusted friendships may be prioritized, which, given compromises potentially introduced in larger networks, may favor smaller networks (Thompson et al., 2024).

Both the theoretical and empirical programs we propose for understanding how individuals manage friendship networks may require nuanced appreciation of the special adaptive problems that friendship networks serve depending on life circumstances. Crucial tests of theoretical predictions about how people manage friendship networks may lie in identifying these differential patterns.

8. Summary

Friendships are highly important, special forms of relationships that individuals have with non-kin, ones characterized by stake-based cooperation and, hence, qualities of communal relationships. Recent modeling finds that relative irreplaceability is a key feature that maintains stakes in friendships. Irreplaceability and, hence, the benefits that accrue to individual friendships may be

affected by connections with other friends. Hence, we stress that people not only require adaptations for developing friendships; they should have adaptations for developing, maintaining, and regulating friendship networks. We have provisionally discussed special adaptive problems that arise in the context of regulating friendship networks and potential solutions. Yet many questions remain, as optimal construction of a friendship network may depend on circumstances. A key overarching question, then, may not be *whether* friendship networks are characterized by a more efficient flow of benefits to self depending on features of a network but, rather, *what are the circumstances* that affect how qualities of networks affect their net benefits. For theory to more precisely specify how individuals can best manage their friendship networks to offer benefits, explicit modeling of friendship networks (e.g., through agent-based modeling of networks) is likely needed. As well, new, ambitious strategies of empirically examining friendships in the context of networks are called for. How people manage their friendship networks likely varies as a function of particular demands of life stages and gender. These variations represent not only a challenge to understand key adaptations for regulating friendship networks; they are also potentially fertile ground for testing conceptual frameworks that concern them.

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